

Firms' Foreign Exchange Hedging

Nicolas Hommel	Thibaut Piquard
University of Chicago	Banque de France

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How do firms respond to exchange rate volatility?

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 - ▶ operate across currencies and countries,
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 - ▶ firms hedge to invest and pay employees even under unfavorable exchange rates.

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How much risk do firms hedge with derivatives and what motivates hedging?

Euro Area as a laboratory

- Main obstacle to empirical work is lack of data
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- Exploit unique European regulation
 - ▶ mandatory daily and contract-level reporting for all Euro Area entities,
 - ▶ complete picture of Eurozone firms' derivatives portfolio.

This paper

- Document key facts at odds with current understanding of currency hedging
 1. firms that face more currency risk hedge a larger fraction of that risk,
 2. trading costs are small for EUR/USD forwards,
 3. EU firms rarely post cash collateral; when they do, it is small,
 4. cash positions are uncorrelated with hedging.

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 4. cash positions are uncorrelated with hedging.
- Develop a quantitative risk management model to explain these facts
 - ▶ standard mechanism = external financing costs & limited borrowing,
 - ▶ new mechanisms = earnings smoothing & hedge adjustment costs.

Main contributions and related literature

1. Empirical importance of currency risk and currency hedging

Adler Dumas '86; Jorion '90; Amihud 94; He Ng '98; Allayannis Ofek '01; Guay Kothari '03; Bodnar et al. '02; Bartram et al. '10; Barbiero '21; Efing et al. '22; Adams Verdelhan '22; Welch Zhou '24; Alfaro et al. '25.

- ▶ Measure currency risk before and after hedging, confirm incomplete hedging.

2. Channels that limit hedging: collateral, discriminatory pricing, CIP deviations

Rampini Viswanathan '10, '11, '25; Du et al. '18; DeMarzo '19; Hau et al. '21; Kubitza et al. '25.

- ▶ Document that hedging is inexpensive and OTC FX markets are efficient.

3. Determinants of hedging

Froot et al. '93; Rampini Viswanathan '10; Bolton et al. '11; Rampini Sufi Viswanathan '14; Bodnar et al. '95 '97; Mian '96; Geczy et al. '97; Froot '01; Allayanis Weston '01; Giambona et al. '18.

- ▶ Build quantitative model to replicate hedging moments and quantify “liquidity channel.”

Outline

Background and Data

Measuring Currency Risk

Four Facts on Currency Hedging

Dynamic Risk Management Model

Conclusion

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Firm-level panel of financial hedging

- **2017–2023.** Daily contract-level reporting under European regulation (EMIR).

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Example contract

- ▶ Reporting counterparty. Deutsche Bank
- ▶ Counterparty side. Seller
- ▶ Other counterparty. Carrefour
- ▶ Contract type. Forward
- ▶ Currency pair. EUR/USD
- ▶ Notional amount. \$1,000,000
- ▶ Forward price. 1.17
- ▶ Maturity date. 2023/07/17
- ▶ Execution timestamp. 2021/09/30 11:07:36 UTC
- ▶ Collateralization. Uncollateralized

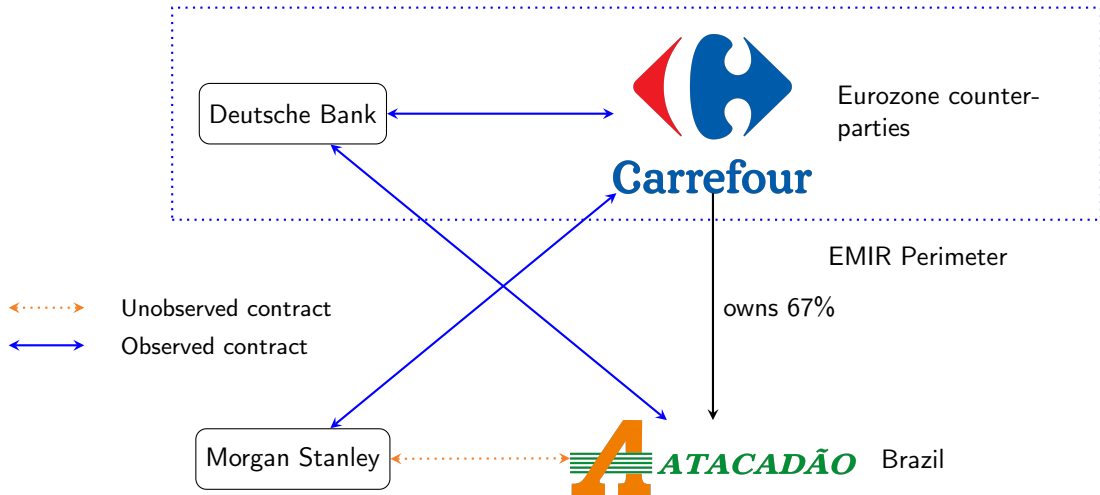
Firm-level panel of financial hedging

- **2017–2023.** Daily contract-level reporting under European regulation (EMIR).
- **2000–2023.** Consolidated quarterly financial statements from Compustat Global
 - ▶ keep firms located in Eurozone, Sweden, Denmark, Norway, and Switzerland,
 - ▶ exclude financial and real-estate firms,
 - ▶ consolidate 10,000 legal entities into Compustat heads for every year.

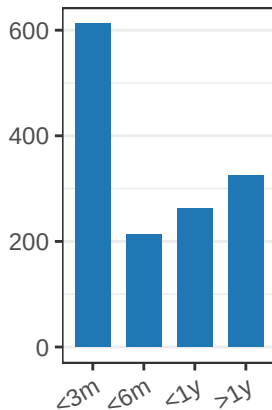
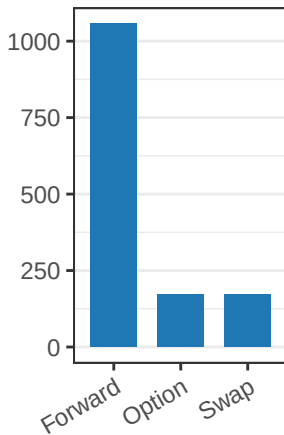
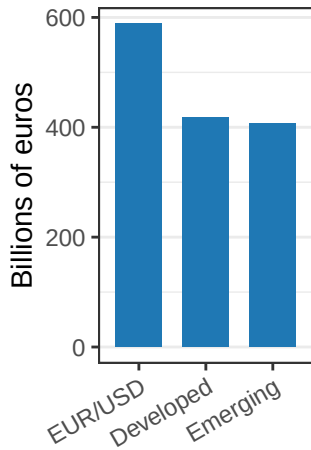
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 - ▶ consolidate 10,000 legal entities into Compustat heads for every year.
- **Final sample.**
 - ▶ 1.5+ million derivatives contracts,
 - ▶ 1,000+ publicly listed firms (after EMIR filters + nonmissing accounting variables),
 - ▶ 4 dealer banks per firm on average.

Data coverage and firm consolidation



Firms mainly trade forwards on developed currencies, across all maturities



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 - ▶ estimate factor model $\neq$ currency mismatch accounting (Adler Dumas '86).

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 1. separate FX risk from non-FX risk,
 2. estimate fraction of FX risk hedged with financial derivatives.

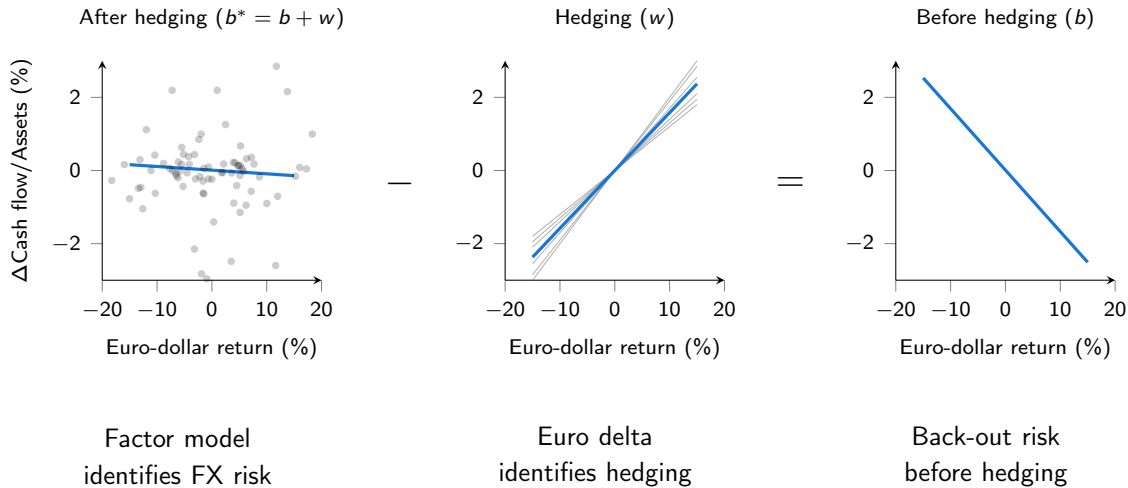
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 - ▶ common approach models profits and FX risk (Bodnar et al. '02; Bartram et al. '10),
 - ▶ requires strong assumptions on demand, costs, and market structure.

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- Our approach = Reconstruct exposure before hedging using derivatives data.

Measuring currency risk before and after hedging



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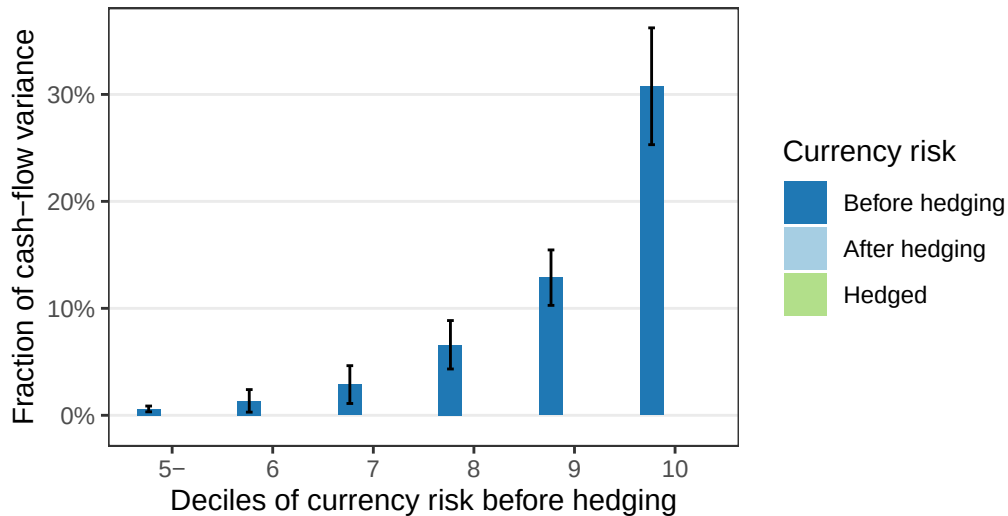
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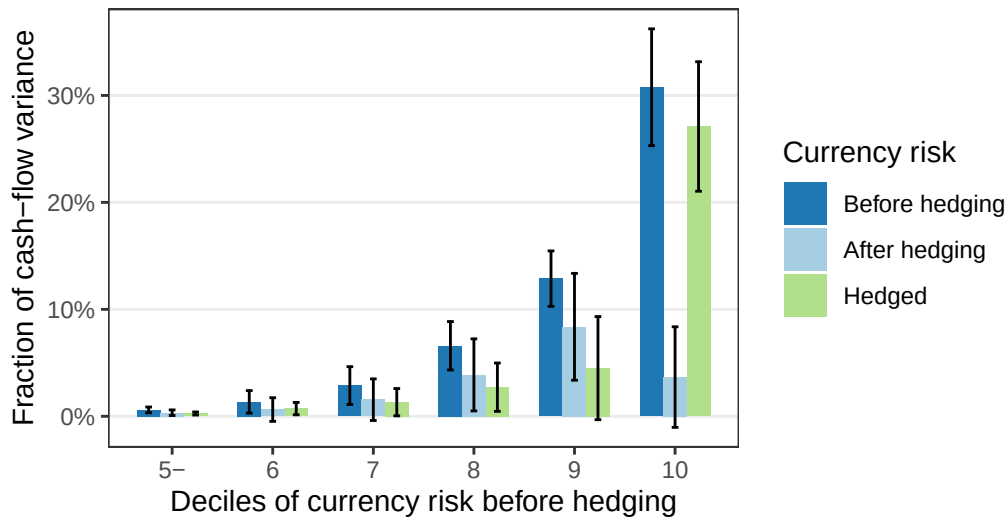
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Fact 1. Firms facing more risk



Fact 1. Firms facing more risk hedge a larger *fraction* of that risk



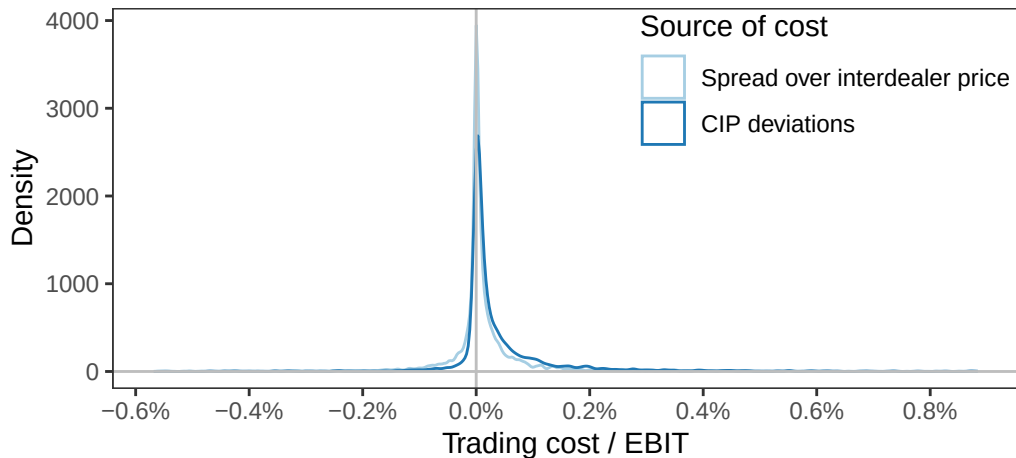
Informing hedging theory

- + **Liquidity channel** (Standard story: Froot Scharfstein Stein '93)
 - ▶ hedging allows firms to finance operations following adverse FX shocks.

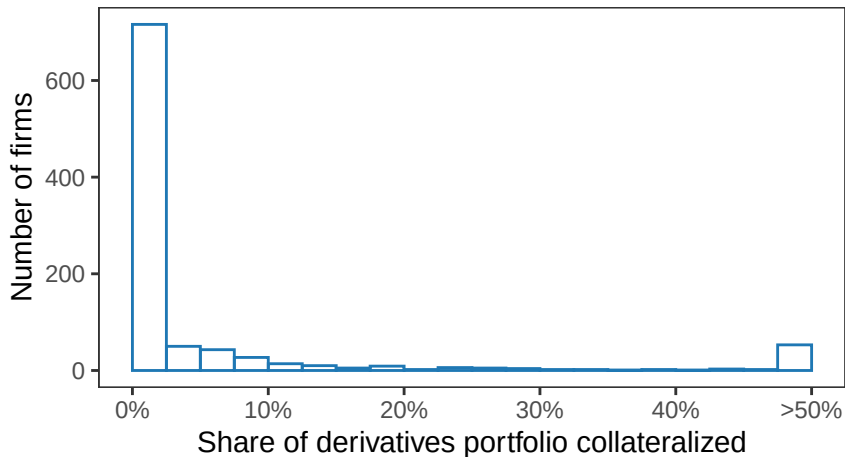
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 - ▶ hedging allows firms to finance operations following adverse FX shocks.
- **Trading costs channel**
 - ▶ dealer banks may charge spread over interdealer price (Hau et al. '21),
 - ▶ prices deviate from CIP (Du et al. '18, Kubitza et al. '25).
- **Collateral channel** (Rampini Viswanathan '10 '11 '25)
 - ▶ hedging is severely limited by collateral constraints.

Fact 2. Trading costs are small relative to EBIT for EUR/USD forwards



Fact 3. EU firms rarely post explicit collateral



Diversification

Commodity companies

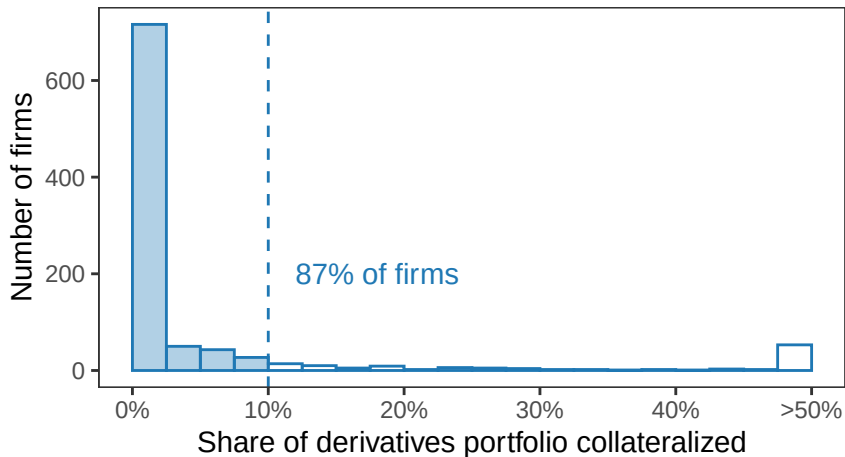
Risk premium

Implicit collateral

Value-at-risk definition

Regulatory exemptions

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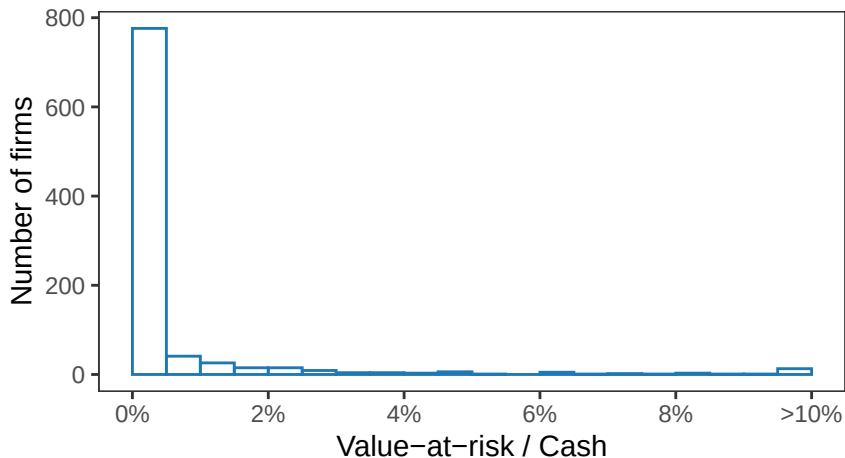
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Fact 3. EU firms rarely post explicit collateral and value-at-risk is small



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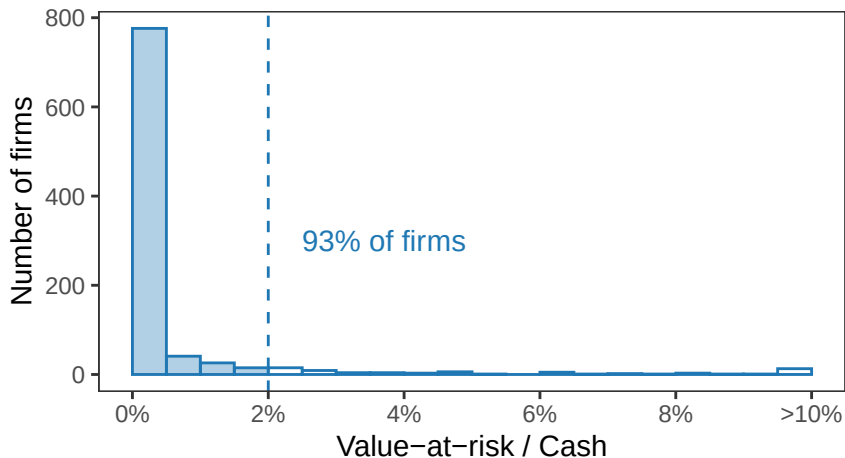
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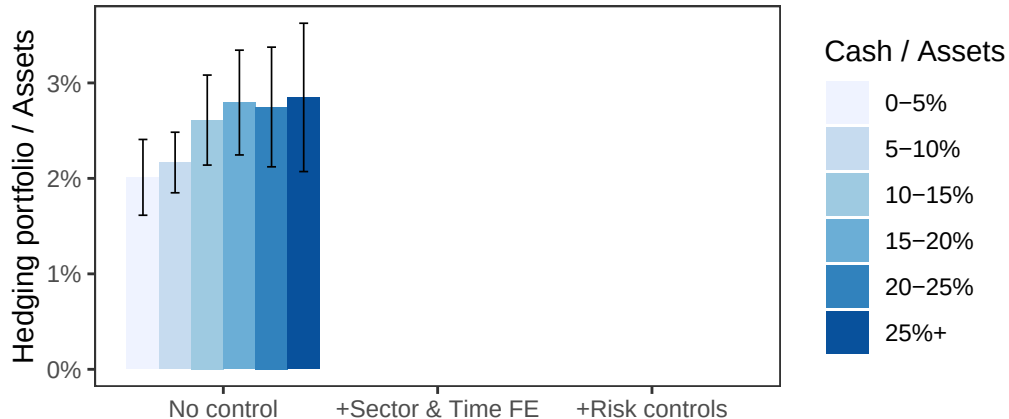
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Fact 4. Cash is uncorrelated with hedging



Standard errors = block bootstrap by firm.

Rare disasters

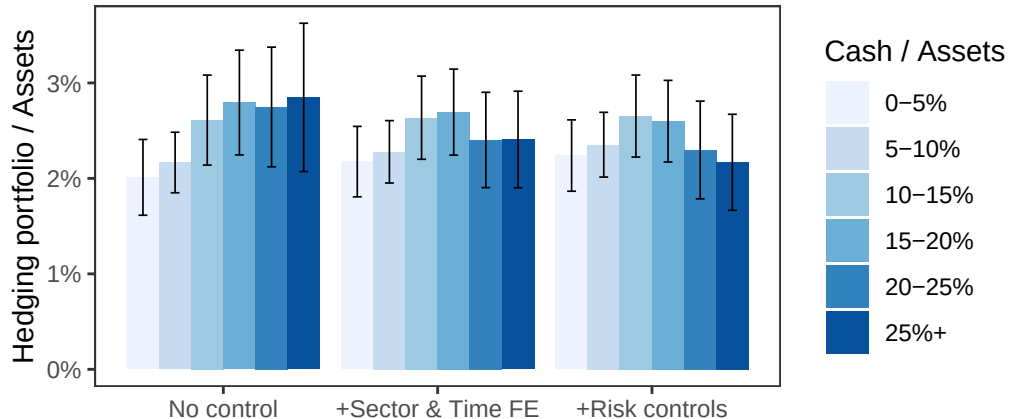
Fraction of risk hedged

Book leverage

Net debt

Payout policy

Fact 4. Cash is uncorrelated with hedging



Standard errors = block bootstrap by firm. Risk controls = EBIT standard deviation.

Rare disasters

Fraction of risk hedged

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Taking stock and next steps

- **Empirical findings.**
 1. Firms facing more risk hedge a larger *fraction* of that risk.
 - 2 & 3. Hedging is inexpensive.
 4. Cash-rich firms hedge the same as others.

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- **Model.** Replicate hedging (Fact 1) given costs (Facts 2 & 3) and liquidity (Fact 4).
 - ▶ How much hedging demand do standard frictions generate?
 - ▶ What, if any, additional mechanisms can match the data?

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- ▶ How much hedging demand do standard frictions generate?

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- **Counterfactuals.**

- ▶ What is the value of hedging FX risk for firms?

- ▶ How do changes in collateral regulation or product availability impact firms' FX hedging?

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Model overview

- **Mechanisms generating hedging** to explain why and how much firms hedge
 1. **Standard “liquidity channel”**
 - borrowing is limited and equity issuance is costly.
 2. **Earnings smoothing** generates hedging demand in all states (Fact 4)
 - e.g., investors learning about fundamentals, managers’ human capital tied to the firm.

Model overview

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 1. Standard “liquidity channel”
 - borrowing is limited and equity issuance is costly.
 2. **Earnings smoothing** generates hedging demand in all states (Fact 4)
 - e.g., investors learning about fundamentals, managers’ human capital tied to the firm.
- **Mechanisms limiting hedging** to explain why firms do not hedge all of their risk
 1. Small trading costs (Fact 2).
 2. No collateralization constraint (Fact 3).
 3. Hedge adjustment costs
 - e.g., governance frictions, hedge accounting on long-term contracts.

Key assumptions

- Exogenous cash-flows.
- Risk-neutral shareholders.
- Three financial instruments:
 1. One period risk-free bonds.
 2. Equity (negative dividends).
 3. FX forwards.

Dynamic financing problem

$$v(s_t) = \sup_{a_t}$$

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$$s_t = (m_t \quad y_t \quad b_t \quad n_t \quad z_t) \quad ; \quad a_t = (b_{t+1} \quad d_t \quad h_t)$$

Exogenous cash-flows.

1. Profit margins w/ FX risk (m).
2. Sales w/ idiosyncratic risk (y).

Three financial instruments.

1. One period risk-free bonds (b).
2. Equity and dividends (d).
3. FX forwards (h, n, z).

Dynamic financing problem

$$v(s_t) = \sup_{a_t} \underbrace{\iota(d_t)}_{\text{Dividend flow}} + \underbrace{\beta \mathbf{E}[v(s_{t+1}) \mid s_t, a_t]}_{\text{Continuation value}} + \underbrace{\varphi(x_t)}_{\text{Earnings smoothing}}$$

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$$\underbrace{\iota(d) = \begin{cases} d & \text{if } d \geq 0 \\ (1 + \lambda)d & \text{if } d < 0 \end{cases}}_{\text{Equity issuance costs}}$$

$$\underbrace{\varphi(x) = \begin{cases} 0 & \text{if } \gamma = 0 \\ (1 - e^{-\gamma x})/\gamma - x & \text{if } \gamma > 0 \end{cases}}_{\text{Smoothing}}$$

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$$d_t = (1 - \tau) \left(\underbrace{y_t m_t}_{\text{EBIT}} \right) - \underbrace{f y_t}_{\text{Other costs}}$$

- y_t = sales, follows geometric AR(1).
- m_t = profit margins = $\alpha_m + \rho_m m_{t-1} + \eta_t$, η_t shock.

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- E_t = exchange rate, follows geometric random walk. User cost Neoclassical

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- b_t = face value of debt with interest rate $r(b_t)$
 - ▶ borrowing limit $b_{\max}$.

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- h_t = new hedges.
- n_t = notional outstanding = $E_t / E_{t-1} \times h_{t-1}$.
- z_t = “blended rate” = $E_{t-1} / E_t \times \kappa_0$, $1 - \kappa_0$ = trading costs.

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- h_t = new hedges, extension with long-term forwards $\delta \leq 1$.
- n_t = notional outstanding = $E_t/E_{t-1} \times (h_{t-1} + (1 - \delta)n_{t-1})$.
- z_t = “blended rate” = $E_{t-1}/E_t \times (\kappa_0 w_{t-1} + z_{t-1}(1 - w_{t-1}))$ (w = weight).

Dynamic financing problem

$$v(s_t) = \sup_{a_t} \iota(d_t) + \beta \mathbf{E}[v(s_{t+1}) \mid s_t, a_t] + \underbrace{\varphi(x_t)}_{\substack{\text{Earnings} \\ \text{smoothing}}}$$

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Calibration and estimation for key parameters

$$\left(\underbrace{\alpha_y \quad \rho_y \quad \sigma_\nu \quad \alpha_m \quad \rho_m \quad \beta_m \quad \sigma_\eta}_{\text{Cash-flow parameters}} \quad \underbrace{f \quad \lambda \quad \kappa_1 \quad \gamma}_{\text{Financial parameters}} \right)$$

- Two steps
 1. calibrate cash-flow parameters externally to match EBIT moments,
 2. estimate financial parameters internally model with SMM.

Calibration and estimation for key parameters

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- Two steps
 1. calibrate cash-flow parameters externally to match EBIT moments,
 2. estimate financial parameters internally model with SMM.
- Equity issuance cost λ and other costs f : identified by payouts and debt.
- Earnings smoothing γ and hedge adjustment costs κ_1 :
 - ▶ intuition = hedging identifies γ/κ_1 , response to short-run cost shock identifies κ_1 ,
 - ▶ moments = hedging in deciles 9/10 + elasticity to CIP (Hacioglu-Hoke et al. '25).

Estimation results: Main parameters

Estimation results: Cash-flow parameters

Estimation method: Cash-flow parameters

Earnings smoothing and hedge adjustments costs match key facts

Moment	Data	Model specification
Hedging Decile 10 (%)	84	
Hedging Decile 9 (%)	37	
CIP semi-elasticity	2	

[Skip to: Conclusion](#)

Earnings smoothing and hedge adjustments costs match key facts

Moment	Data	Model specification
		Liquidity
Hedging Decile 10 (%)	84	26
Hedging Decile 9 (%)	37	22
CIP semi-elasticity	2	755

[Skip to: Conclusion](#)

Earnings smoothing and hedge adjustments costs match key facts

Moment	Data	Model specification	
		Liquidity	Baseline
Hedging Decile 10 (%)	84	26	73
Hedging Decile 9 (%)	37	22	39
CIP semi-elasticity	2	755	2

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Earnings smoothing and hedge adjustments costs match key facts

Moment	Data	Model specification		
		Liquidity	Baseline	Smoothing
Hedging Decile 10 (%)	84	26	73	59
Hedging Decile 9 (%)	37	22	39	42
CIP semi-elasticity	2	755	2	823

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Earnings smoothing and hedge adjustments costs match key facts

Moment	Data	Model specification			
		Liquidity	Baseline	Smoothing	Adjustment
Hedging Decile 10 (%)	84	26	73	59	2
Hedging Decile 9 (%)	37	22	39	42	0
CIP semi-elasticity	2	755	2	823	44

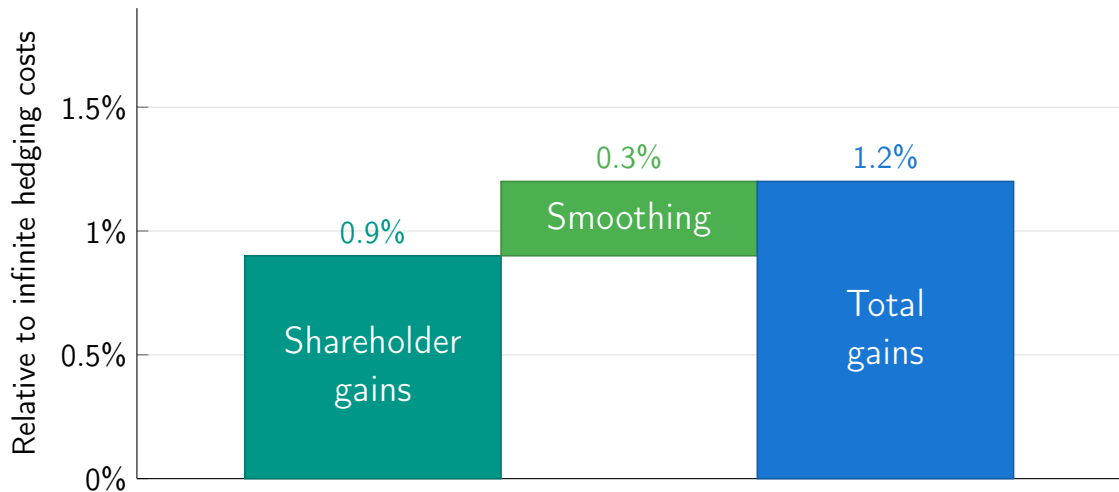
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Hedging Decile 9 (%)	37	22	39	42	0
CIP semi-elasticity	2	755	2	823	44
Debt and hedging corr.	≤ 0	0.72	-0.33	0.06	0.53

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Model implies gains from hedging on the order of 1% of firm value [Details](#)



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Background and Data

Measuring Currency Risk

Four Facts on Currency Hedging

Dynamic Risk Management Model

Conclusion

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Conclusion

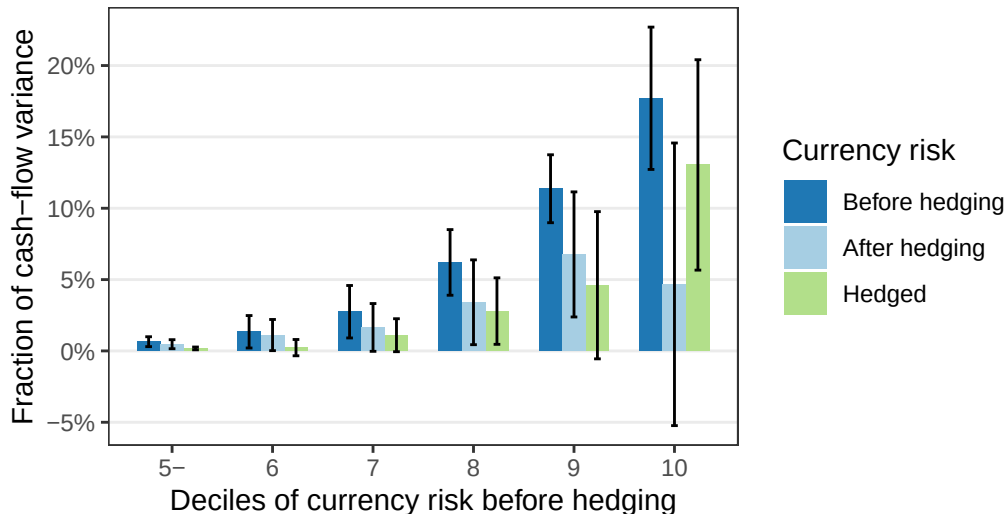
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- Broader questions
 - ▶ firms = impact of hedging on product market pricing?
 - ▶ financial intermediaries = limits on supply of foreign exchange derivatives?

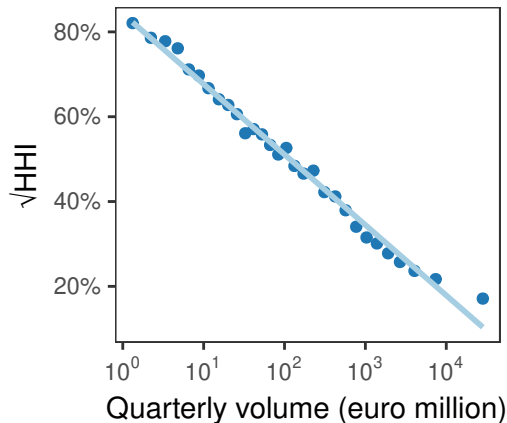
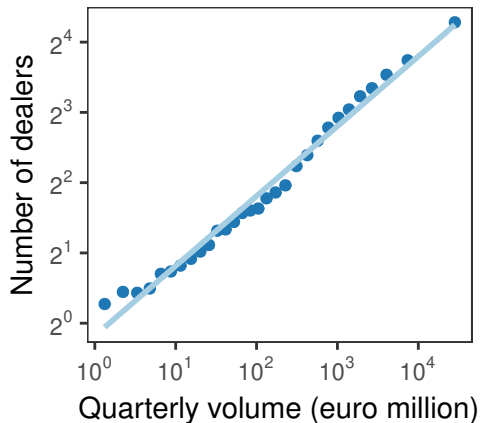
Appendix

Robustness of risk estimation to using pretax profits instead of EBIT



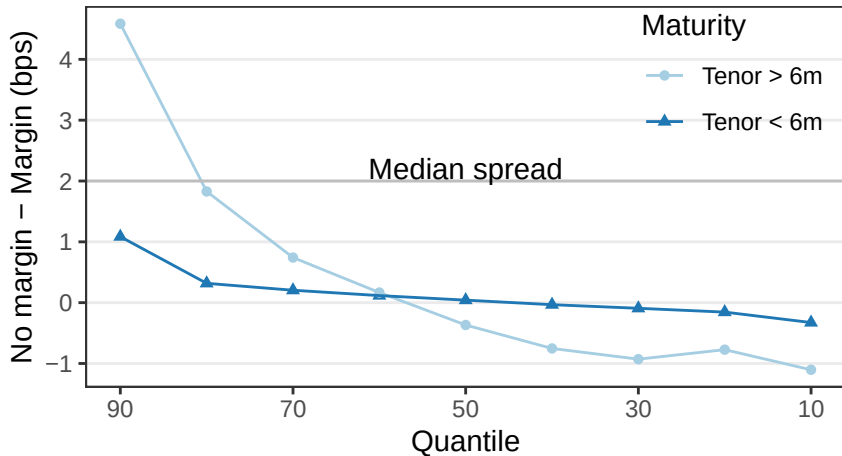
High-volume firms use more dealers and spread trades across dealers more

EUR/USD contracts

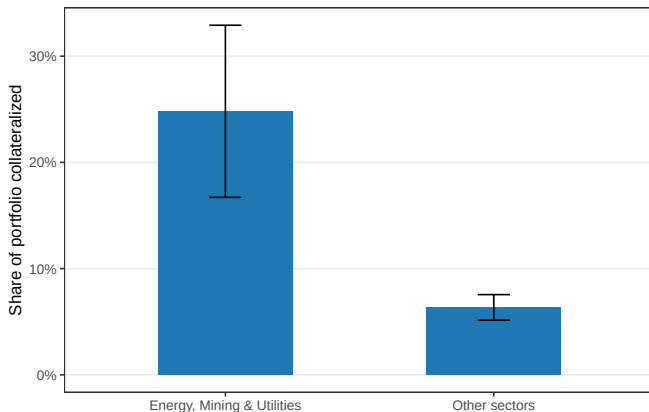


Markup quantiles are larger for long-term uncollateralized contracts

EUR/USD forwards



Energy, mining, and utility companies post more collateral



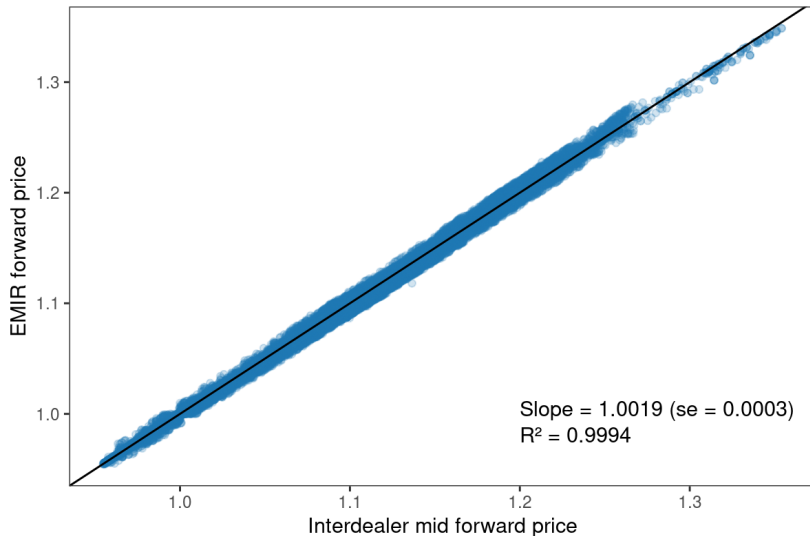
Time series of collateralization status

Definition of distance to interdealer price [Back](#)

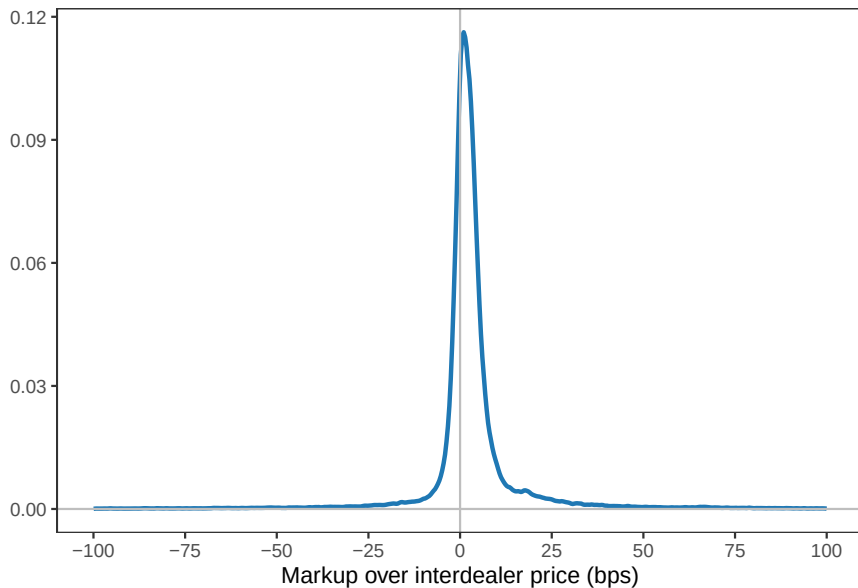
1. Combine EMIR forward prices with interdealer quotes from LSEG Tick History
 - ▶ interdealer spot and forward premium for tenors 1m, 2m, 3m, 6m, 1y, at high frequency,
 - ▶ forward price $F_{ct}(T)$ for contract c at timestamp t and tenor T .
2. Interpolate tenors linearly to get interdealer bid $B_t^*(T)$ and ask $A_t^*(T)$.
3. Define log markup over interdealer price (including bid-ask spread)

$$m_{ct}(T) = \begin{cases} \log F_{ct}(T) - \log B_t^*(T) & \text{if long} \\ \log A_t^*(T) - \log F_{ct}(T) & \text{if short} \end{cases} .$$

Forward prices reported in EMIR align with high quality interdealer data [Back](#)



Distribution of markups over interdealer price across contracts [Back](#)



Definition of value-at-risk

- Value-at-risk from a 1 in 25 years margin call given observed collateral

1. compute the euro delta exposed across currencies k

$$\text{Total euro delta}_i = \frac{1}{T} \sum_{kt} \text{Euro delta}_{itk},$$

2. define normalized value-at-risk as

$$\text{Normalized value-at-risk}_i = \frac{\text{Total euro delta}_i \times \text{Collateralization share}_i}{\text{Cash and equivalents}_i}.$$

- Assumptions

1. all currency pairs are simultaneously hit by an unfavorable exchange rate shock,
2. firms can net out their exposures across counterparties.

Fact 3. EU firms rarely post explicit collateral and value-at-risk is small

Two possible interpretations for Fact 3

- Fact 3 consistent with
 1. Collateral constraints do not bind for FX hedging.
 2. Implicit collateral constraints may bind + No explicit collateral in equilibrium.

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Two possible interpretations for Fact 3

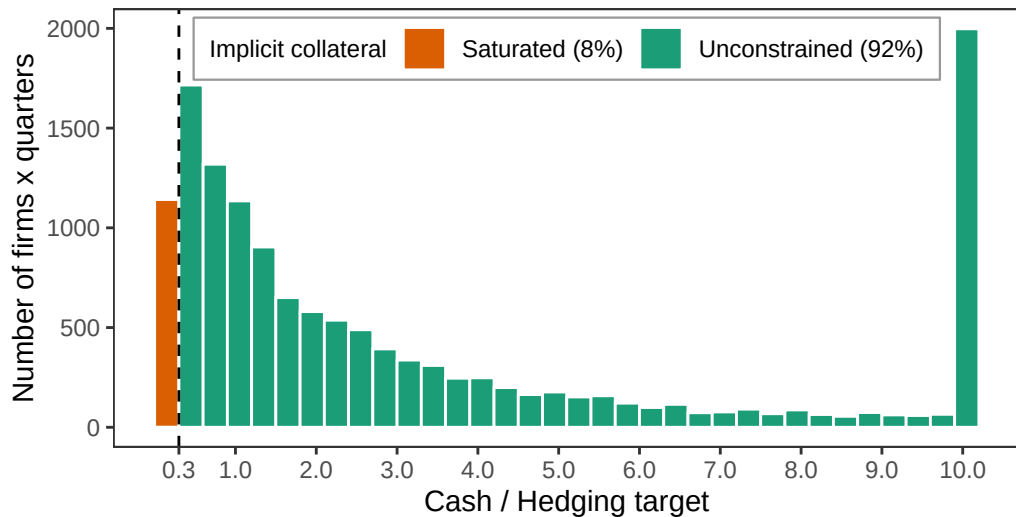
- Fact 3 consistent with
 1. Collateral constraints do not bind for FX hedging.
 2. Implicit collateral constraints may bind + No explicit collateral in equilibrium.
- Posit firms need $\$ \lambda$ in cash to hedge $\$1$

$$\frac{\text{Cash and equivalents}}{|\text{Hedging target}|} \geq \lambda.$$

- ▶ observe cash and equivalents and hedging target (currency exposure before hedging),
 - ▶ value-at-risk suggests $\lambda < 0.3 \Rightarrow$ take conservative $\lambda = 0.3$.
- How many firms are constrained given data and $\lambda = 0.3$?

Fact 3. Firms rarely post explicit collateral and value-at-risk is small

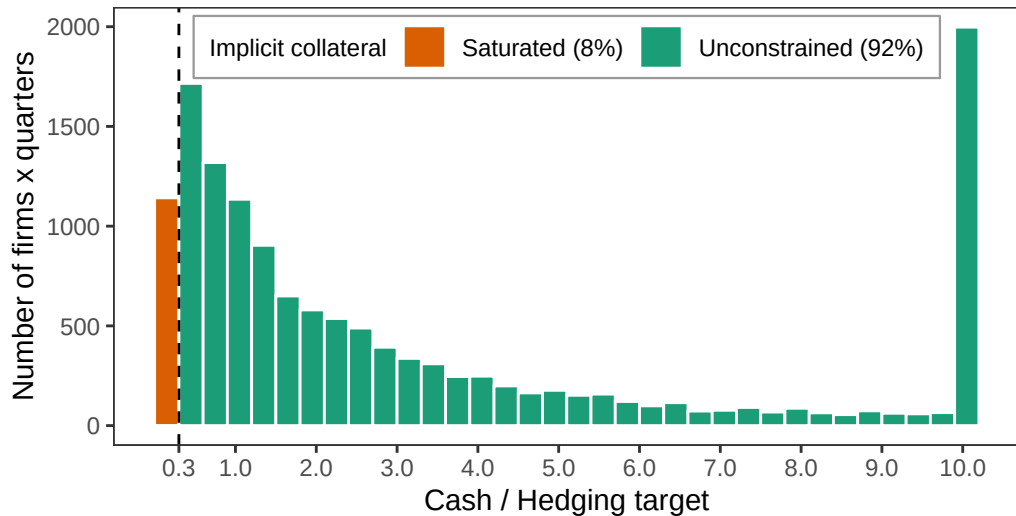
Calibrated model implies that implicit collateral constraint is often slack



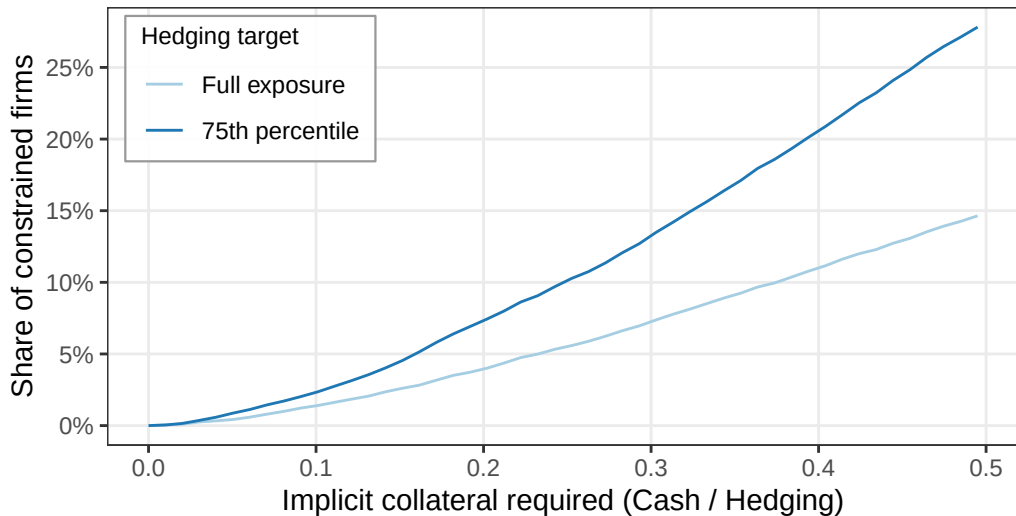
Additional results: Implicit collateral (1/4) [Back](#)

1. Baseline calibration. [To results](#)
2. Robustness to alternative calibration. [To results](#)
3. Robustness to estimating constraints from the data. [To results](#)

Additional results: Implicit collateral baseline (2/4) [Back](#)

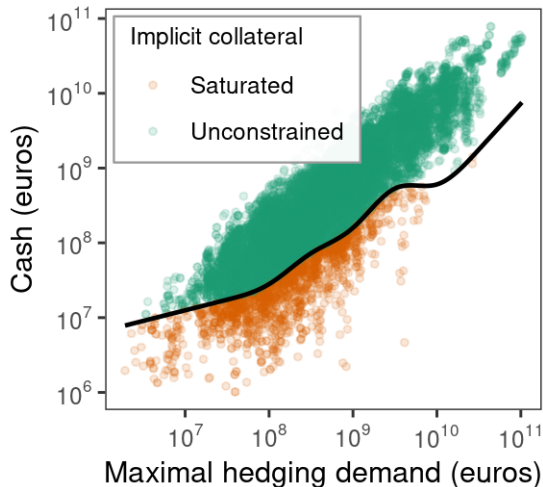
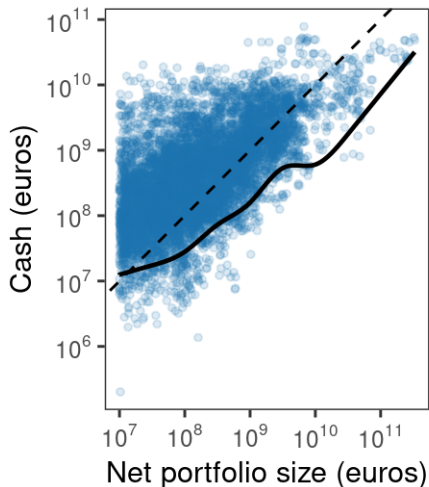


Additional results: Share constrained under alternative implicit collateral calibrations (3/4) [Back](#)



Additional results: Estimating upper bound on implicit collateral requirements (4/4)

[Back](#)

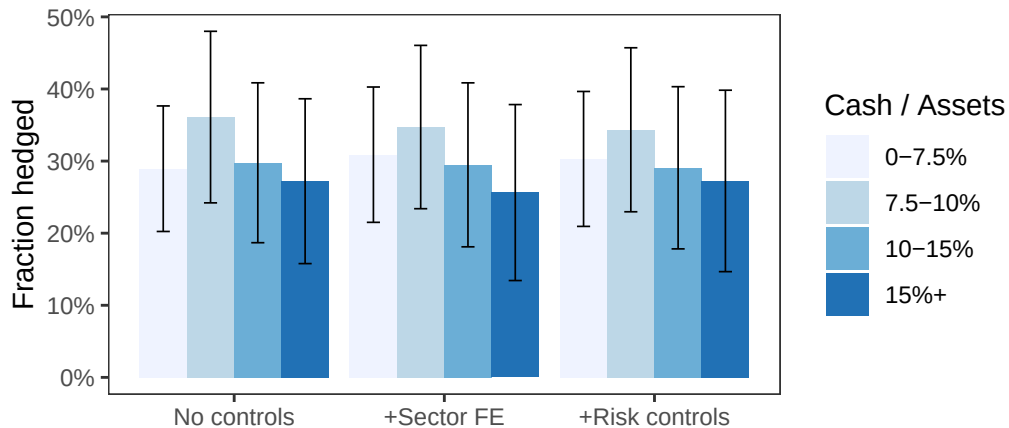


Policymakers are careful to exempt nonfinancial hedging from regulation

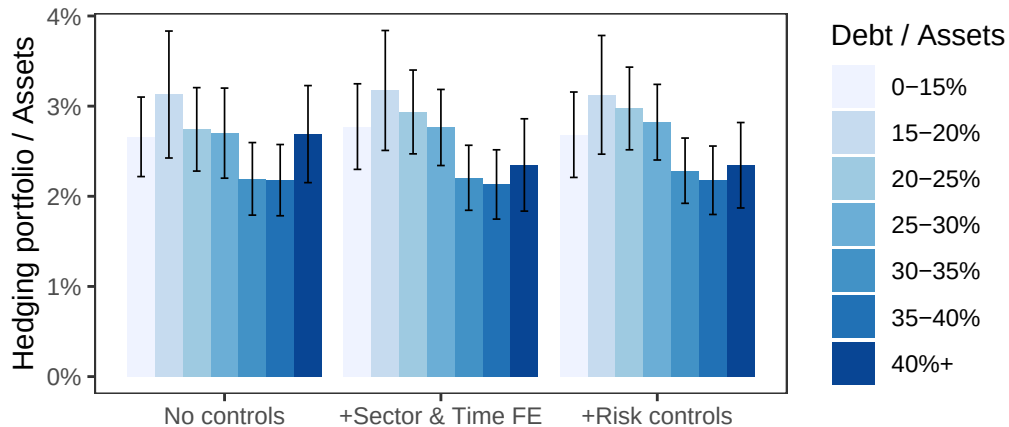
Derivatives are an important tool businesses use to manage costs and market volatility. Congress recognized that imposing clearing and exchange trading requirement on commercial end-users could raise transaction costs when there is a substantial public interest in keeping such costs low.

Back

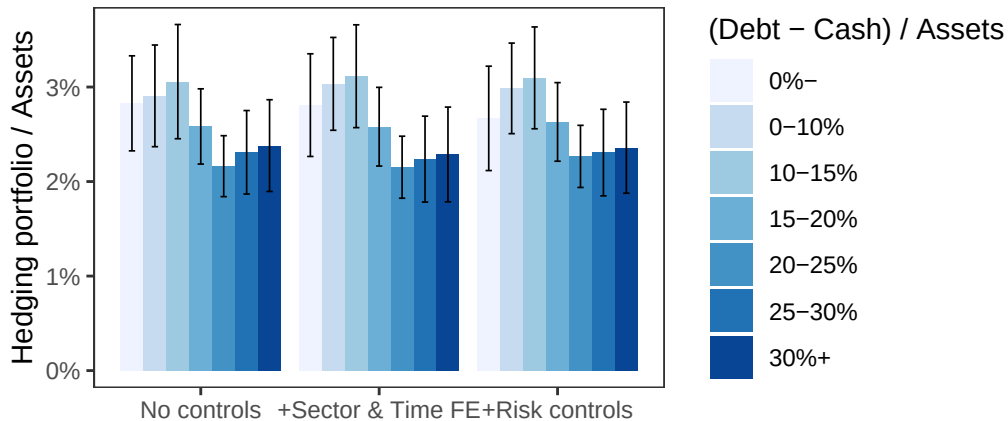
Cash is uncorrelated with fraction of risk hedged [Back](#)



Firms with high leverage don't hedge more [Back](#)



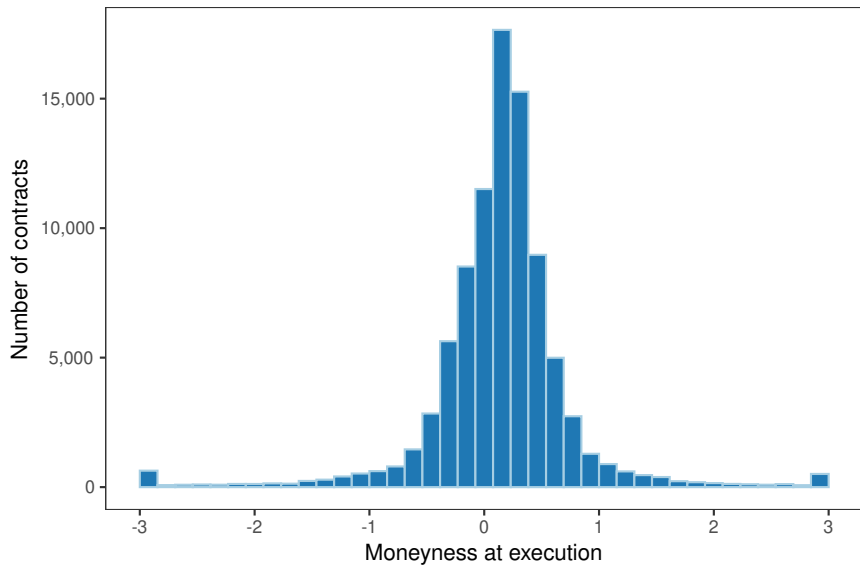
Firms with high net leverage don't hedge more [Back](#)



Issuing equity and buying back stocks don't predict hedging [Back](#)

	Euro delta for EUR/USD		
	(1)	(2)	(3)
Equity issuance indicator	-0.29 (0.36)	-0.44 (0.37)	-0.41 (0.44)
Stock buyback indicator	-0.29 (0.34)	-0.35 (0.34)	-0.22 (0.33)
Observations	15,343	14,110	14,077
Sector and Time fixed effects	—	✓	✓
Risk controls	—	—	✓

Firms trade options near-the-money [Back](#)



Computing euro deltas from individual contracts

- Use contract delta as local measure of sensitivity
 - ▶ use ± 1 for forwards and swaps,
 - ▶ use Black–Scholes for options, assuming they are European vanilla options.
- “Euro delta” measures the firm’s portfolio exposure (Tufano ’96)

$$\underbrace{H_{itk}}_{\text{Euro delta}} = \underbrace{\sum_c \delta_{ckt} \times \text{Notional}_{ckt}}_{\text{Volume-weighted delta}}$$

- ▶ contract c , currency k , time t , firm i ,
- ▶ use contracts that expire within 360 days.

Defining optimal path-independent derivatives loadings

- Model: Fixing initial $P(0)$ and final $P(T)$ exchange rates, profits and losses Π are

$$\Pi_T - \Pi_0 \simeq \sum_{t=0}^{T-1} (P_{t+1} - P_t) X_t = \underbrace{(P_T - P_0) \bar{X}_T}_{\text{Path-independent}} + \underbrace{\text{Cov}(P_{t+1} - P_t, X_t - \bar{X}_T)}_{\text{Path-dependent}},$$

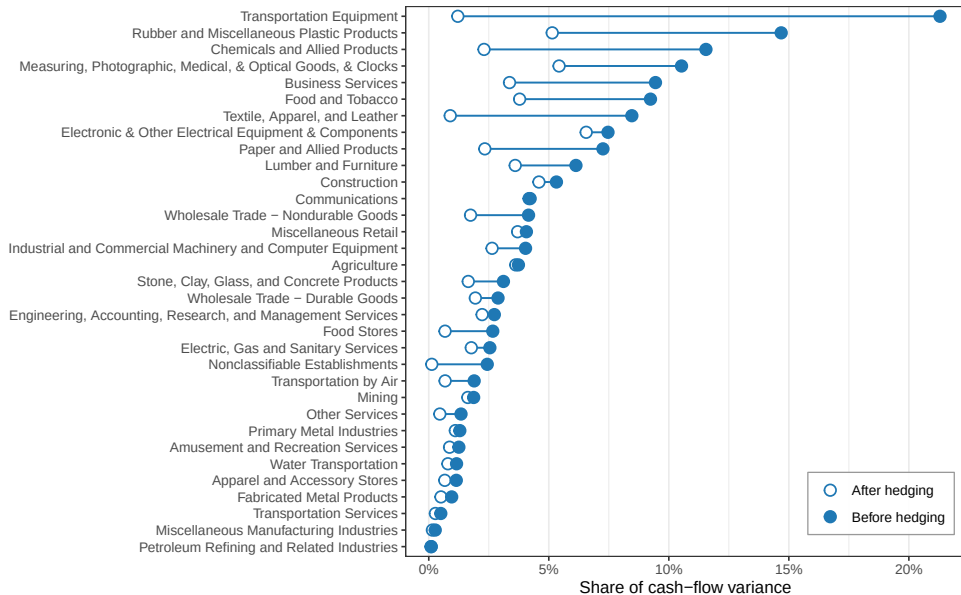
where X is the euro delta.

- Theoretically: Average euro delta is **best path-independent approximation**.

Average portfolio maturity is under one year in most industries Back



Currency risk before and after hedging by sector

[Back](#)


- User cost model microfoundations cash-flow process

$$\max_K AK^\alpha - rK \propto A^{\frac{1}{1-\alpha}} r^{-\frac{\alpha}{1-\alpha}}.$$

- ▶ identical functional form, for example: $A^{1/(1-\alpha)} \leftarrow m - f$ and $r^{-\alpha/(1-\alpha)} \leftarrow y$.
- Assumes financial constraints are slack for investment problem.

- In [neoclassical investment model](#) with adjustment costs and random productivity
 - ▶ strong positive correlation between investment and productivity z ,
 - ▶ exchange rate shocks hit productivity z .
- In the data, [investment and exchange rates are uncorrelated](#) $\Rightarrow$ counterfactual.

Calibration and estimation for key parameters: Cash-flow parameters

- Invert cash-flow process in closed form to **match empirical moments**
 - ▶ EBIT moments: average, autocorrelation, standard deviation, currency risk.
 - ▶ sales-to-assets moments: autocorrelation, volatility.
- Estimate **two sets of parameters**
 1. high currency risk targets currency risk of 32% of EBIT variance,
 2. low currency risk targets currency risk of 15% of EBIT variance.

Description	Symbol	Lower risk	Higher risk
Sales process			
Persistence	ρ_y	0.88	0.88
Idiosyncratic volatility	σ_ν	0.18	0.18
Drift (%)	α_y	-0.86	-0.86
Profit margin process			
Persistence	ρ_m	0.55	0.55
FX exposure	β_m	0.20	0.29
Idiosyncratic volatility (%)	σ_η	1.37	0.88
Drift (%)	α_m	1.83	1.86

- Compare firm value in **two counterfactual scenarios**
 1. baseline calibration,
 2. infinite hedging costs.
- Decompose firm value to **isolate gains to shareholders**

$$\mathbf{E}v = \underbrace{\frac{\mathbf{E}\iota(d)}{1-\beta}}_{\text{Shareholder value}} + \underbrace{\frac{\mathbf{E}\varphi(x)}{1-\beta}}_{\text{Smoothing value}},$$

where we compute expectations under the infinite hedging costs stationary distribution.

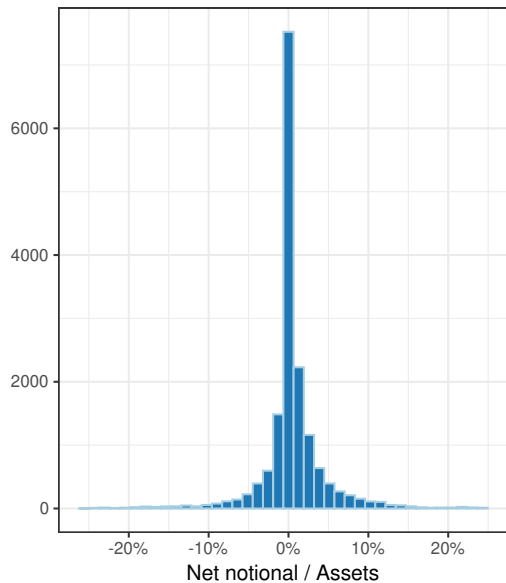
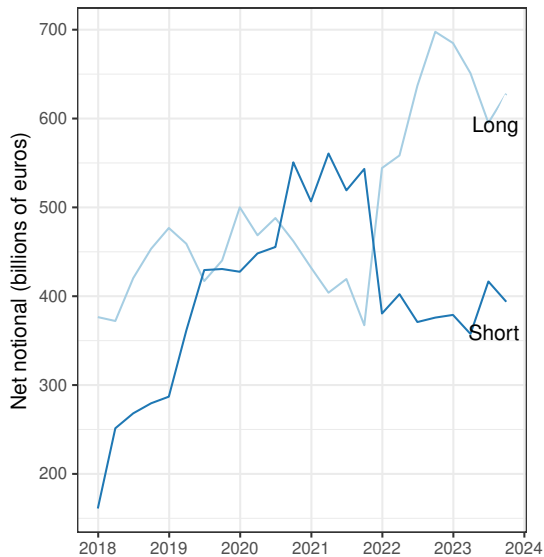
Company	Assets ^a	FX exposure	σ_h^b	$\overline{\text{ROA}}$	$\sigma(\text{ROA})$	σ_π^c	Variance share
Airbus	€131b	+\$22.4b	€2.2b	2.1%	1.9%	€2.5b	50%
Air France	€38b	-\$2.8b	€0.3b	0.6%	5.1%	€1.9b	2%
BMW	€278b	+€33.8b	€2.1b	5.9%	1.7%	€4.8b	16%
Lufthansa	€50b	-\$9.2b	€0.5b	2.0%	5.6%	€2.8b	10%
Ryanair	€17b	-\$4.9b	€0.5b	8.2%	5.0%	€0.9b	23%
Safran	€56b	+\$10.5b	€1.0b	5.6%	3.0%	€1.7b	25%

^a Measured end of 2023.

^b $\sigma_h = \text{Exposure} \times \text{Yearly volatility}$.

^c $\sigma_\pi = \text{Assets}_{2023} \times \sigma(\text{ROA})$.

Descriptive statistics on EUR/USD hedging [Back](#)



Estimating currency risk loadings in three steps

1. Estimate factor model for currency exposure after hedging

$$\frac{\text{EBIT}_{it} - \text{EBIT}_{it-4}}{\text{Assets}_{it-4}} = b_i^* \cdot f_t + \varepsilon_{it}$$

- ▶ b_i^* currency exposure after hedging and f_t yearly change in exchange rate
- ▶ two-step ridge to limit overfitting given 60 observations/firm: industry then firm.

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3. Back-out currency exposure before hedging $b_i = b_i^* - w_i$.